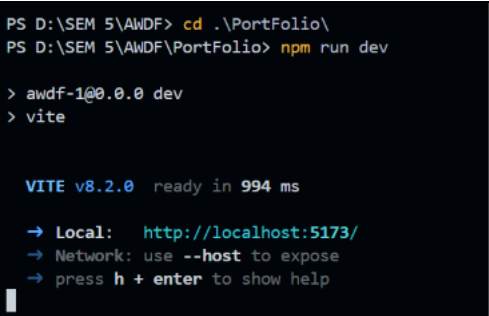


Practical 1: Introduction to React and Component Architecture

Objective	To set up a React development environment using Vite and build a static UI using independently structured, reusable components.
Problem Statement	• Create a React application using Vite for a student portfolio page. • The application must include at least 4 reusable components: Header, About, Skills, and Footer. • Each component must be independently structured and composed into a single-page layout. • No logic or JSX should be duplicated across components. • Props must be used to pass at least one piece of data into a minimum of 2 components.
Project Structure	<pre> student-portfolio/ ├── src/ │ ├── App.jsx │ └── components/ │ ├── Header.jsx │ ├── About.jsx │ ├── Skills.jsx │ └── Footer.jsx </pre>
Implementation	 <pre> PS D:\SEM 5\AWDF> cd .\PortFolio\ PS D:\SEM 5\AWDF\PortFolio> npm run dev > awdf-1@0.0.0 dev > vite VITE v8.2.0 ready in 994 ms → Local: http://localhost:5173/ → Network: use --host to expose → press h + enter to show help </pre> To Create folder with react and JavaScript write the following command: <pre> npm create vite@latest student-portfolio -- --template react cd student-portfolio npm install npm run dev </pre>



Header.jsx:

```
function Header({ name, role }) {
  return (
    <header className="flex min-h-screen w-full items-center justify-center px-4 py-0 text-center">
      <div className="w-full max-w-5xl rounded-2xl border border-sky-300/30 bg-slate-900/70 p-8 shadow-2xl backdrop-blur-sm">
        <p className="m-0 text-xs uppercase tracking-[0.18em] text-slate-300">Student Portfolio</p>
        <h1 className="my-2 text-4xl font-bold text-slate-50 md:text-6xl">{name}</h1>
        <p className="m-0 text-lg font-semibold text-cyan-300">{role}</p>
      </div>
    </header>
  )
}

export default Header
```

About.jsx:

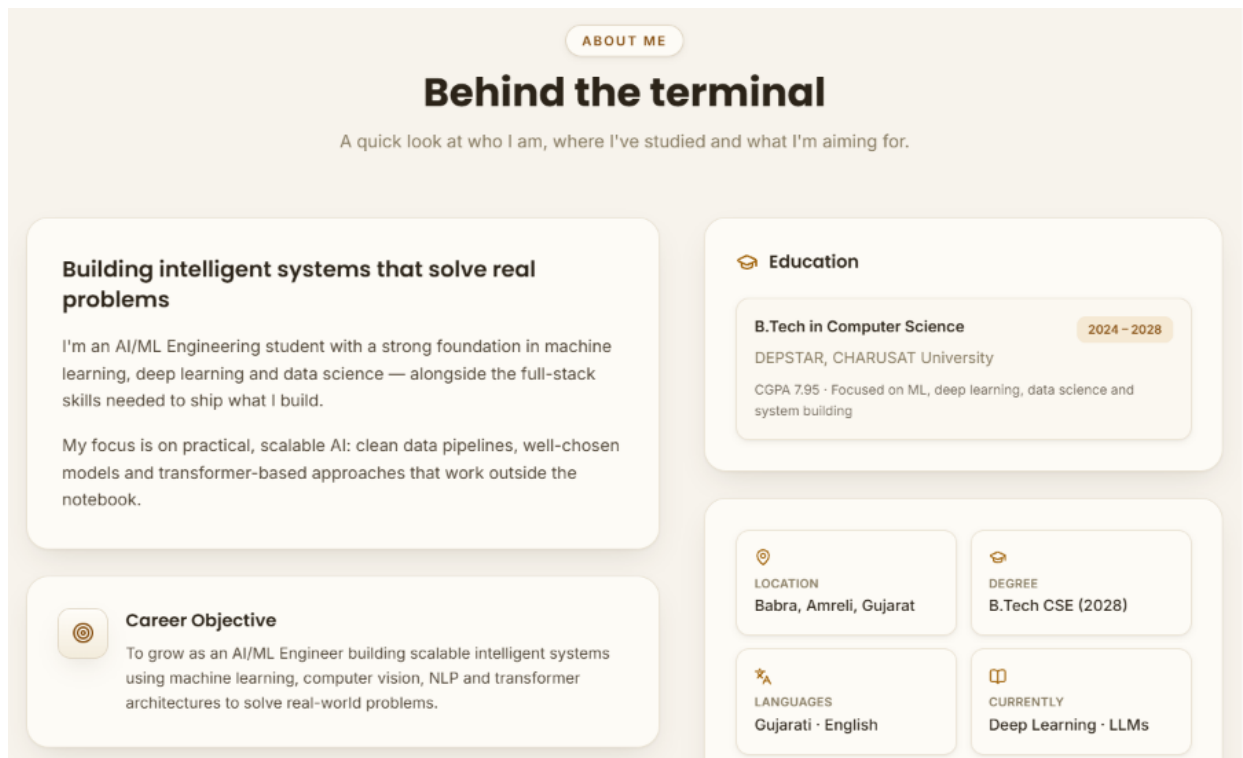
```
function About({ bio, education }) {
  return (
    <section
      className="rounded-2xl border border-sky-300/30 bg-slate-900/70 p-6 shadow-2xl backdrop-blur-sm"
      aria-labelledby="about-title"
    >
      <h2 id="about-title" className="text-2xl font-semibold text-slate-100">
        About Me
      </h2>
      <div className="mt-4 space-y-3">
```

```

    <p className="rounded-xl border border-slate-600/60 bg-slate-800/70 p-4
text-slate-200">{bio}</p>
    <p className="rounded-xl border border-cyan-400/40 bg-slate-800/70 p-4
text-slate-300">{education}</p>
  </div>
</section>
)
}

export default About

```



Skills.jsx:

```

function Skills({ skills }) {
  return (
    <section
      className="rounded-2xl border border-sky-300/30 bg-slate-900/70
p-6 shadow-2xl backdrop-blur-sm"
      aria-labelledby="skills-title"
    >
      <h2 id="skills-title" className="text-2xl font-semibold text-slate-100">
        Skills
      </h2>
      <ul className="mt-4 grid list-none grid-cols-2 gap-3 p-0 md:grid-cols-3">
        {skills.map((skill) => (
          <li
            key={skill}
            className="rounded-xl border border-cyan-300/40 bg-slate-800/80 px-3
py-3 text-center font-semibold text-cyan-100"
          >

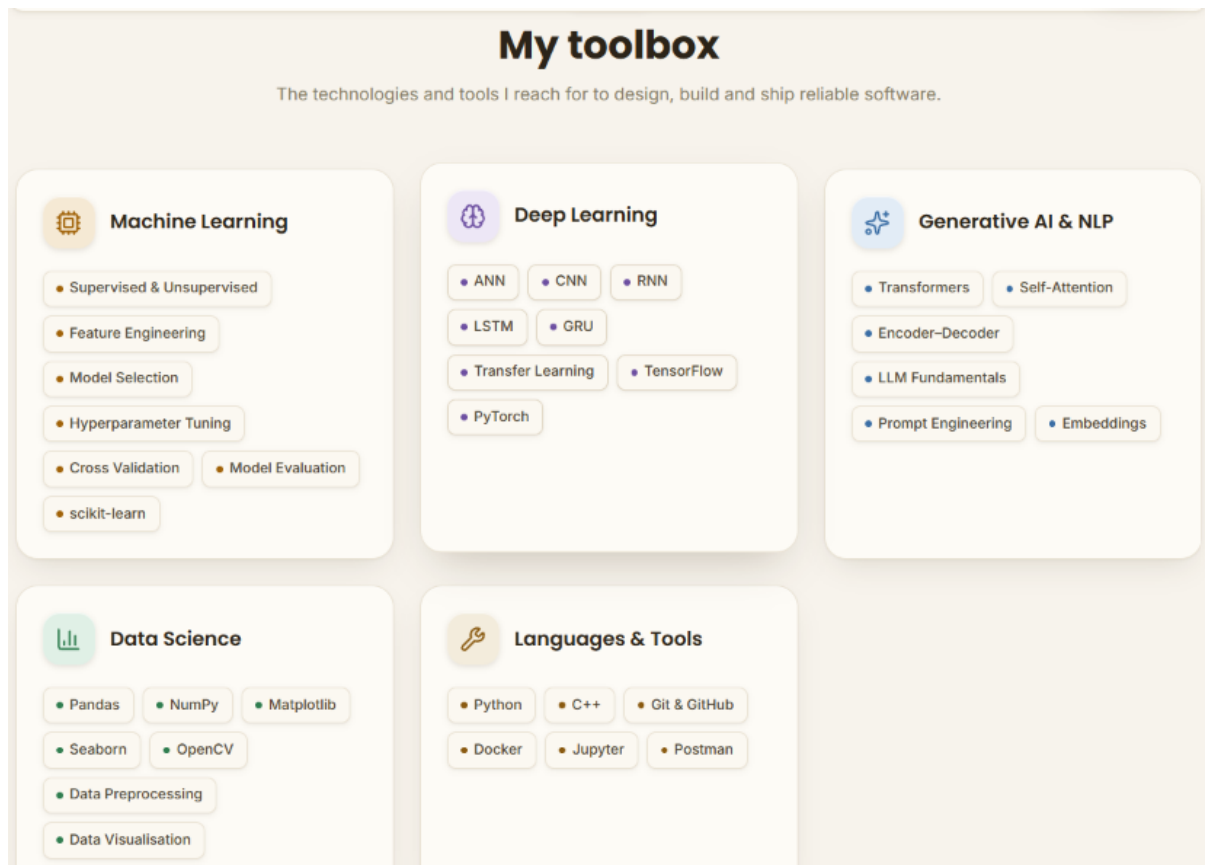
```

```

        {skill}
      </li>
    ))}
  </ul>
</section>
)
}

export default Skills

```



Footer.jsx:

```

function Footer({ name, year }) {
  return (
    <footer className="mt-auto w-full border-t border-sky-300/30 bg-slate-950/70">
      <div className="mx-auto w-full max-w-5xl px-4 py-4 text-center text-sm text-slate-300">
        <p className="m-0">
          © {year} {name}. Built with React + Vite.
        </p>
      </div>
    </footer>
  )
}

export default Footer

```

ST Talaviya Sarthak

AI/ML Engineer · Deep Learning Enthusiast · LLM & NLP Explorer



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App.jsx:

```
import Header from './components/Header'
import About from './components/About'
import Skills from './components/Skills'
import Footer from './components/Footer'

function App() {
  const portfolio = {
    name: 'Sarthak Talaviya',
    role: 'B.Tech Student | Frontend Enthusiast',
    bio: 'I am a third-year engineering student passionate about creating clean and useful web interfaces. I enjoy solving practical problems with React and learning new technologies through projects.',
    education: 'B.Tech in Computer Science, Expected Graduation: 2028',
    skills: ['HTML5', 'CSS3', 'JavaScript', 'React', 'Vite', 'Git'],
  }

  const currentYear = new Date().getFullYear()

  return (
    <div className="min-h-screen flex flex-col">
      <Header name={portfolio.name} role={portfolio.role} />
      <main className="mx-auto w-full max-w-5xl space-y-4 px-4 py-8">
        <About bio={portfolio.bio} education={portfolio.education} />
        <Skills skills={portfolio.skills} />
      </main>
      <Footer name={portfolio.name} year={currentYear} />
    </div>
  )
}

export default App
```

Output

GitHub Repo along with commands:

```
git init
git add .
git commit -m "Initial portfolio with Header, About, Skills, Footer components"
git remote add origin https://github.com/Talaviya-Sarthak/AWDF.git
git branch -M main
git push -u origin main
```

Conclusion

Through this practical, a React development environment was successfully set up using Vite, and a working portfolio UI was built using four independently structured, reusable components. Props

	were used to pass dynamic data (name and skillList) into child components, reinforcing the core React concepts of component composition and unidirectional data flow.
--	---